

```
===== Data Overview =====
```

```
Shape: (891, 12)
```

```
First 5 rows:
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

4			Allen, Mr. William Henry	ma
	Parch		Ticket	Fare Cabin Embarked
0	0		A/5 21171	7.2500 NaN S
1	0		PC 17599	71.2833 C85 C
2	0	STON/O2.	3101282	7.9250 NaN S
3	0		113803	53.1000 C123 S
4	0		373450	8.0500 NaN S

Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

Descriptive Statistics:

	PassengerId	Survived	Pclass	Name	Sex	\
count	891.000000	891.000000	891.000000	891	891	
unique	NaN	NaN	NaN	891	2	
top	NaN	NaN	NaN	Dooley, Mr. Patrick	male	
freq	NaN	NaN	NaN	1	577	
mean	446.000000	0.383838	2.308642	NaN	NaN	
std	257.353842	0.486592	0.836071	NaN	NaN	
min	1.000000	0.000000	1.000000	NaN	NaN	
25%	223.500000	0.000000	2.000000	NaN	NaN	
50%	446.000000	0.000000	3.000000	NaN	NaN	
75%	668.500000	1.000000	3.000000	NaN	NaN	
max	891.000000	1.000000	3.000000	NaN	NaN	

	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
count	714.000000	891.000000	891.000000	891	891.000000	204	889
unique	NaN	NaN	NaN	681	NaN	147	3
top	NaN	NaN	NaN	347082	NaN	G6	S
freq	NaN	NaN	NaN	7	NaN	4	644
mean	29.699118	0.523008	0.381594	NaN	32.204208	NaN	NaN
std	14.526497	1.102743	0.806057	NaN	49.693429	NaN	NaN
min	0.420000	0.000000	0.000000	NaN	0.000000	NaN	NaN
25%	20.125000	0.000000	0.000000	NaN	7.910400	NaN	NaN
50%	28.000000	0.000000	0.000000	NaN	14.454200	NaN	NaN
75%	38.000000	1.000000	0.000000	NaN	31.000000	NaN	NaN
max	80.000000	8.000000	6.000000	NaN	512.329200	NaN	NaN

```
max      80.000000      8.000000      0.000000      NaN      512.529200      NaN      NaN
```

```
Missing Values:
```

```
PassengerId      0
```

```
Survived         0
```

```
Pclass          0
```

```
Name            0
```

```
Sex              0
```

```
Age             177
```

```
SibSp           0
```

```
Parch           0
```

```
Ticket          0
```

```
Fare            0
```

```
Cabin           687
```

```
Embarked        2
```

```
dtype: int64
```

```
==== Survival Analysis ====
```

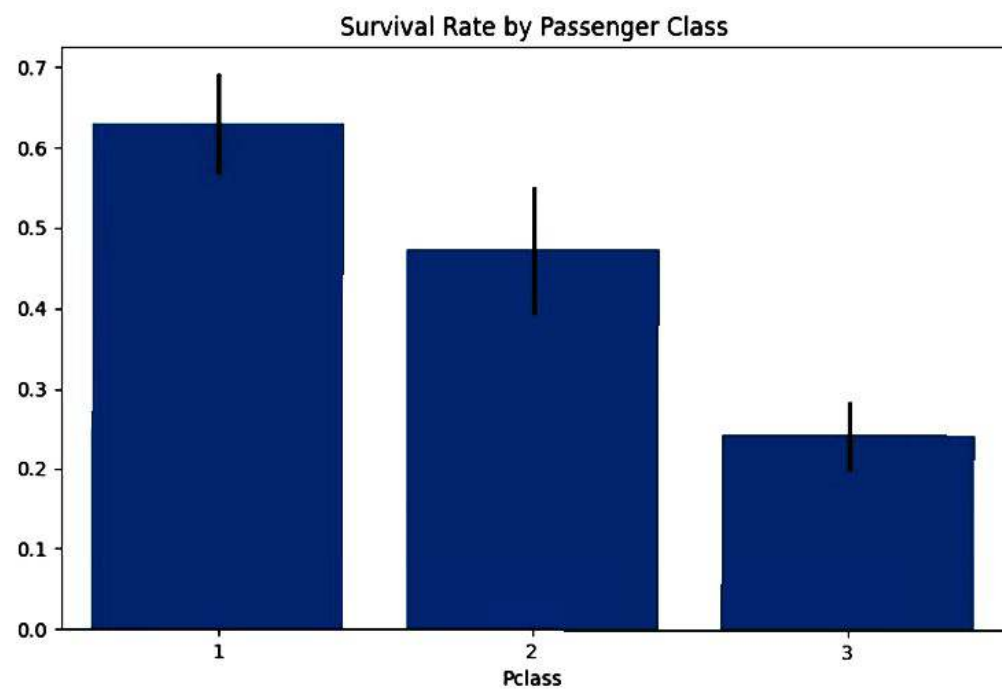
```
Overall Survival Rate: 38.38%
```

```
<ipython-input-4-82e85c47
```

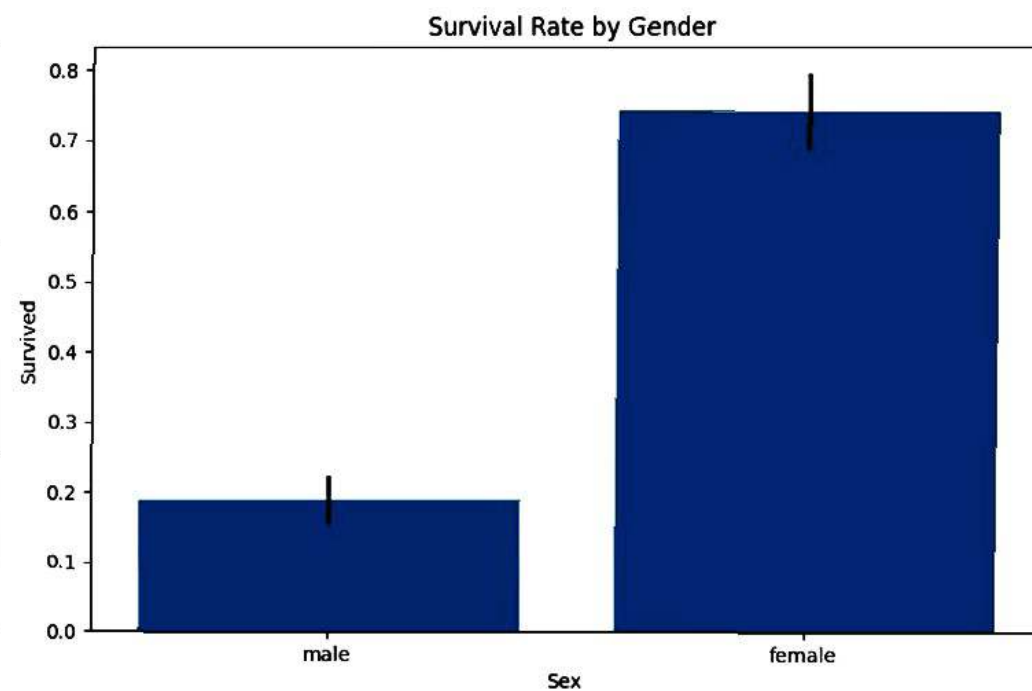
```
The behavior will change
```

Drag from top and touch the back button to exit full screen.

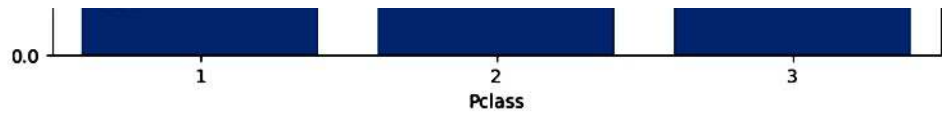
trying to be set on a copy of a
ll never work because the inte



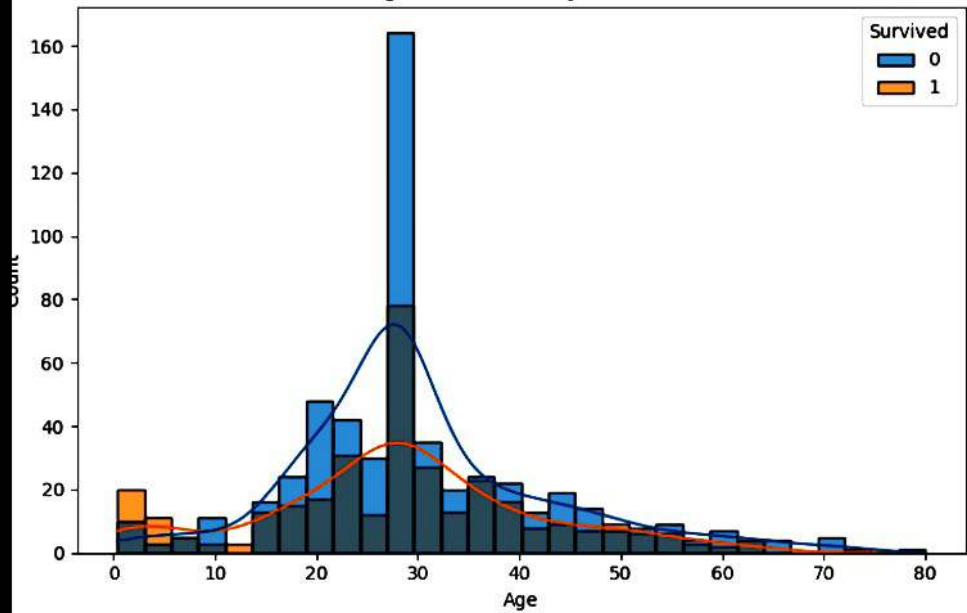
Age Distribution by Survival



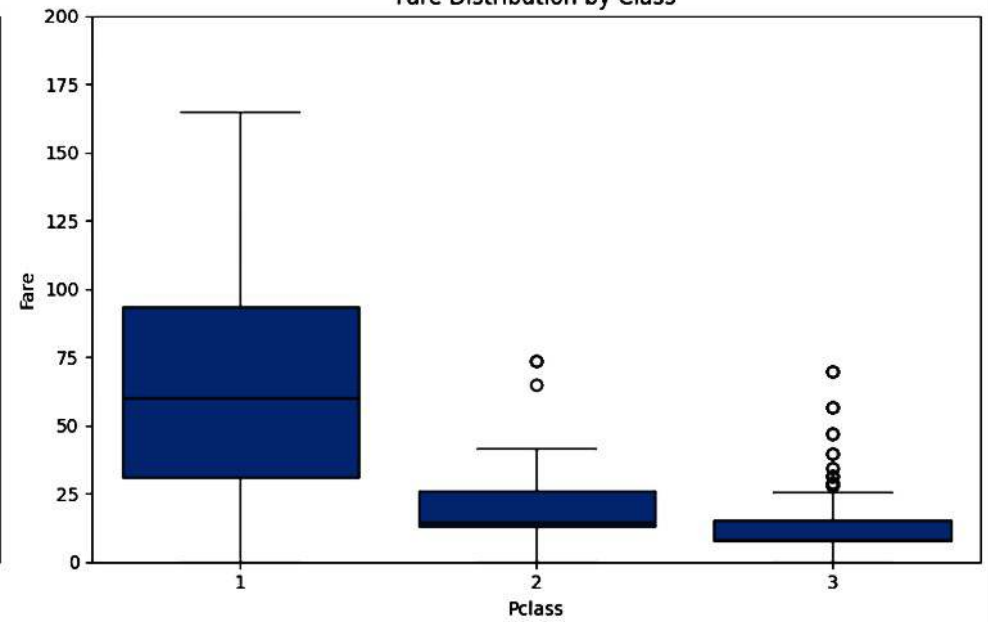
Fare Distribution by Class



Age Distribution by Survival



Fare Distribution by Class



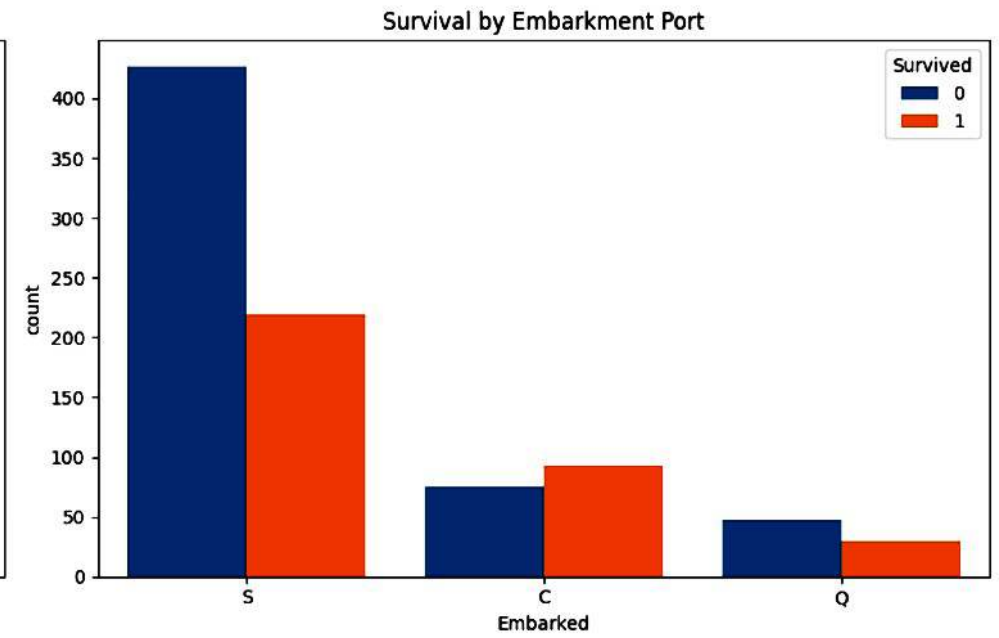
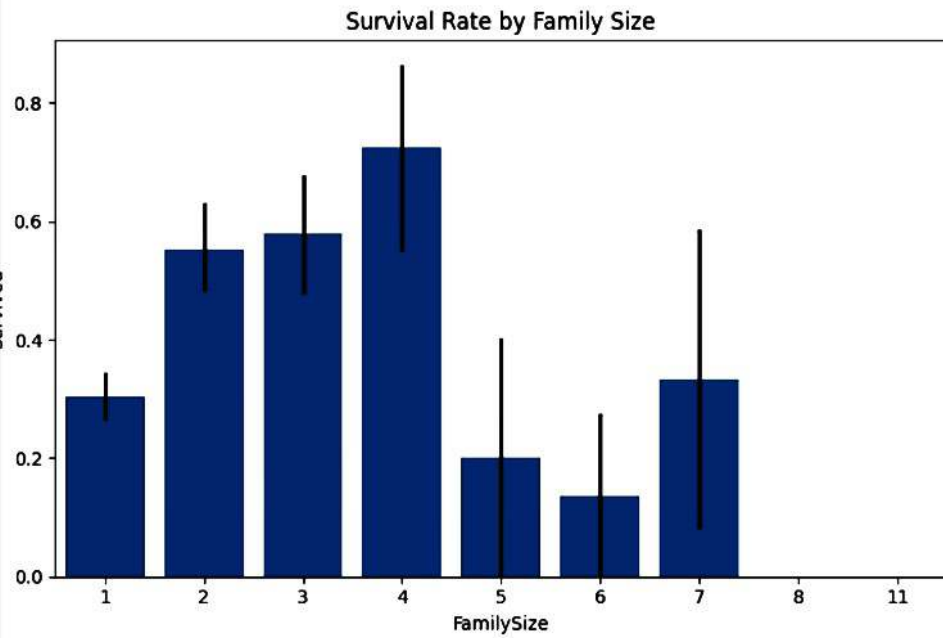
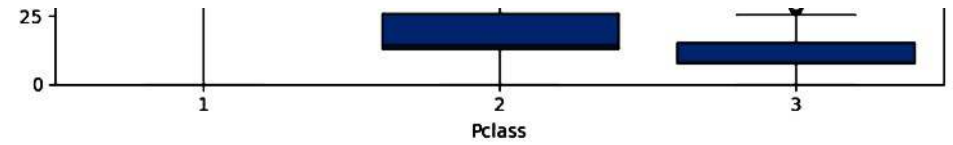
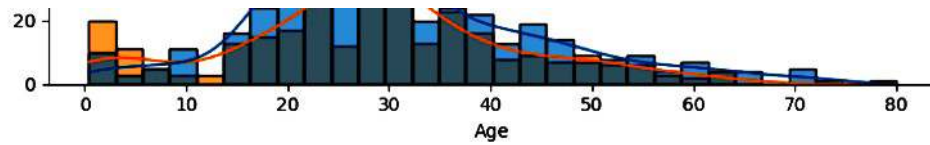
Survival Rate by Family Size



Survival by Embarkment Port

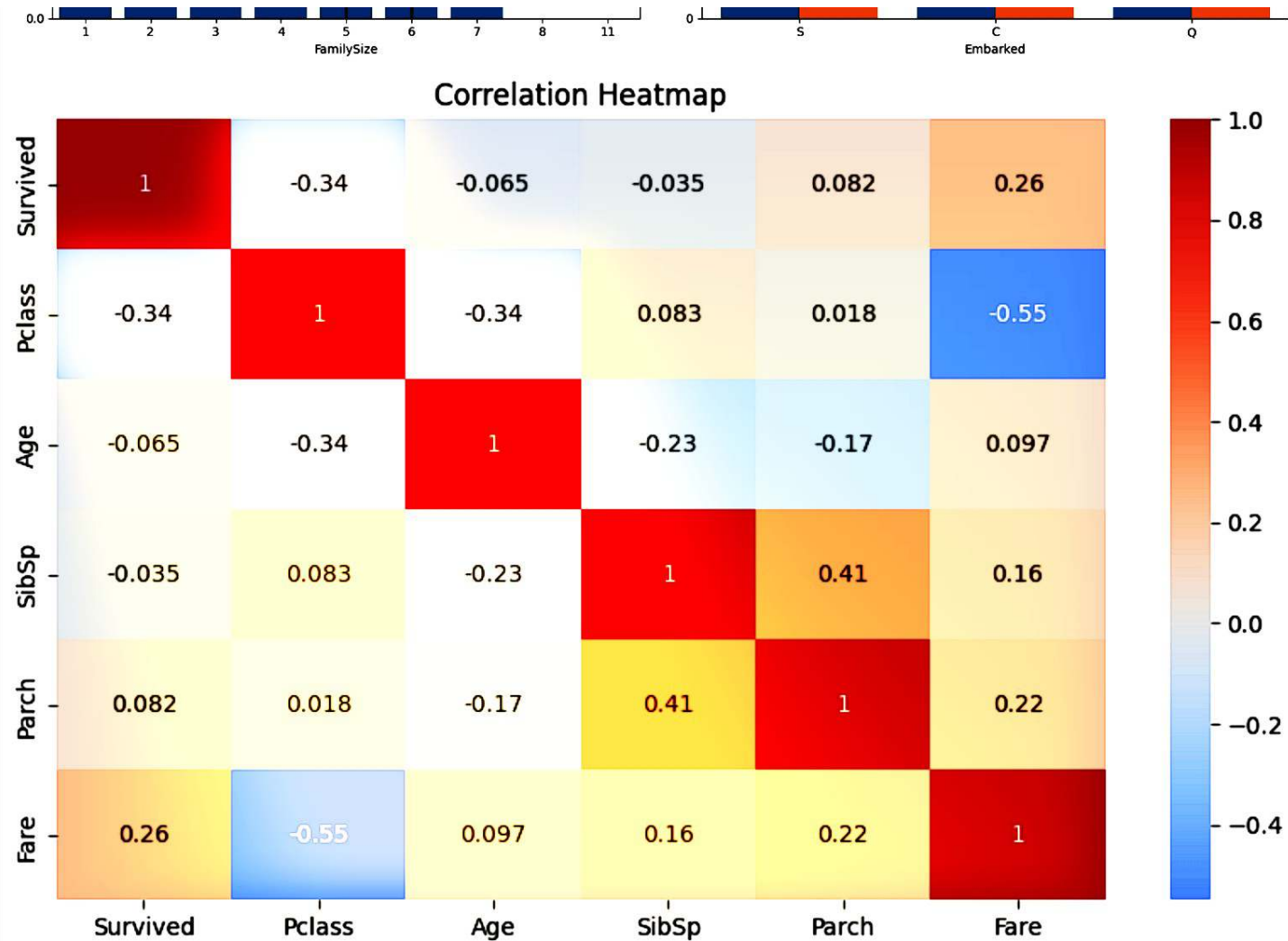


Drag from top and touch the back button to exit full screen.



Drag from top and touch the back button to exit full screen.

1.0



==== Key Findings =====

1. Overall survival rate: 38.38%
2. Female survival rate: 74.20%
3. Male survival rate: 18.89%
4. 1st class survival rate: 62.96%
5. 3rd class survival rate: 24.24%
6. Children (<10) had higher survival rates than adults
7. Higher fare passengers had better survival chances
8. Passengers with 1-3 family members had highest survival rates